
Data Science Project Report — *your project title here*

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Abstract

TODO: one-paragraph summary — problem, method, headline result.

1 Introduction

TODO: introduce the problem, motivation, and the input/output of your task.

2 Related work

TODO: survey prior work. Cite with natbib, e.g. Hastie et al. [2009] proposed ..., and similar tasks were studied by Chicco and Jurman [2020].

3 Dataset and features

TODO: describe the data, the splits, preprocessing, and the features.

4 Methods

$$\text{softmax}(\mathbf{z})_k = \frac{e^{z_k}}{\sum_{j=1}^K e^{z_j}}, \quad k = 1, \dots, K. \quad (1)$$

TODO: describe and motivate the algorithms you chose.

5 Results

$$\text{F1} = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}. \quad (2)$$

TODO: present and discuss your results, referencing every table and figure. For instance, see Table 1 and Figure 1.

Table 1: Test-set performance on the synthetic heart-failure cohort ($n = 360$ held-out patients). All metrics lie in $[0, 1]$, higher is better; **bold** marks the best model per column. AUC is the area under the ROC curve.

Model	Accuracy	Precision	Recall	F1	AUC
Logistic regression	0.769	0.735	0.669	0.700	0.838
Random forest	0.875	0.862	0.821	0.841	0.936
Gradient boosting	0.869	0.860	0.807	0.833	0.919

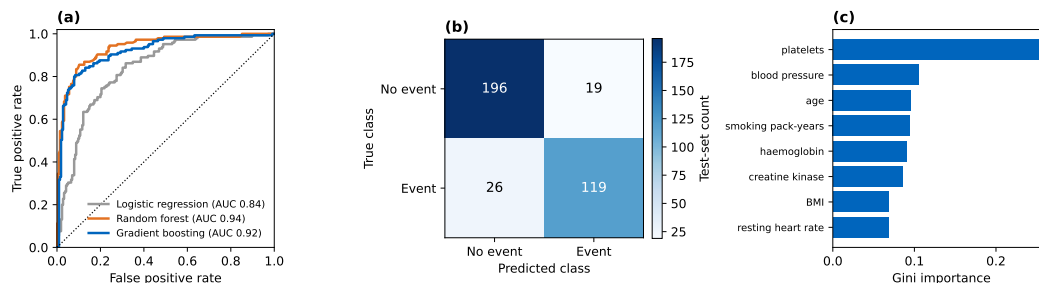


Figure 1: Evaluation of three classifiers on the held-out test set. **(a)** ROC curves; colour encodes the model and the legend reports each AUC. **(b)** Confusion matrix of the best model (random forest); cell colour and number give the test-set count. **(c)** The eight most important features of the random forest (Gini importance).

6 Conclusion

TODO: summarize findings and outline future work.

Contributions

TODO: per-member contributions.

References

- Davide Chicco and Giuseppe Jurman. Machine learning can predict survival of patients with heart failure from serum creatinine and ejection fraction alone. *BMC Medical Informatics and Decision Making*, 20(1):16, 2020. doi: 10.1186/s12911-020-1023-5.
- Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer, New York, 2 edition, 2009. doi: 10.1007/978-0-387-84858-7.

A Appendix

TODO: supporting material, e.g. the hyperparameter table below.

Table 2: Hyperparameters used for the final models.

Model	Hyperparameter	Value
Random forest	n_estimators	300
Random forest	random_state	0
Gradient boosting	learning_rate	0.1